

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

1750–2024

Global Trajectory • Country Totals • Regional Rebalancing
Per Capita Inequality • 2024 Annual Change • Policy Implications

Source: NASA Climate; Our World in Data (Global Carbon Budget 2025)

Atmospheric CO₂ Has Reached 429 ppm

A level unseen in human civilization

What changed?

- Latest Mauna Loa reading (Feb 2026): 429 ppm
- Pre-industrial baseline: ~280 ppm
- Increase exceeds 50% in ~2 centuries
- Primarily linked to fossil fuel combustion

Rate matters:

Comparable natural shifts took millennia, not decades.

Atmospheric CO₂

429 ppm

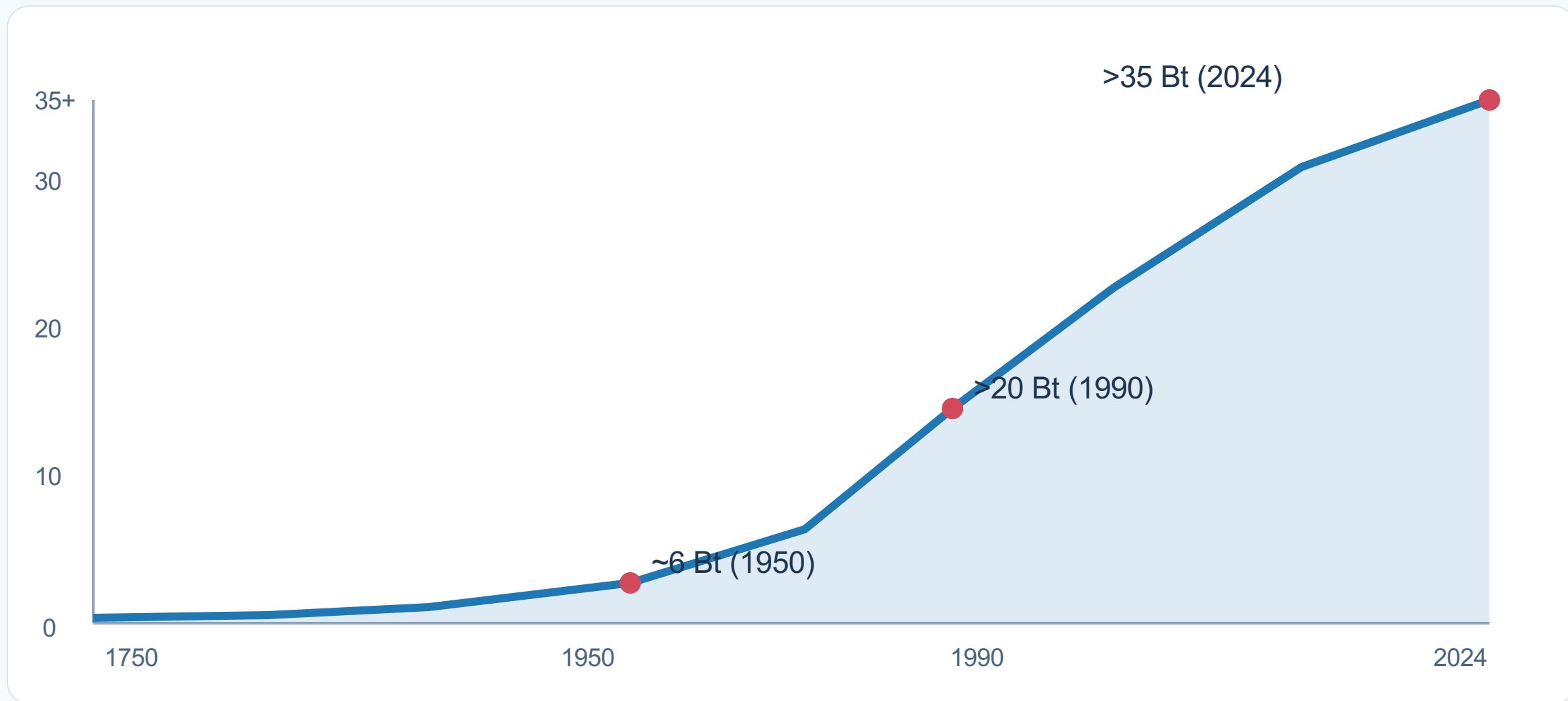
vs. ~280 ppm pre-industrial



>50% increase

From Near-Zero to Over 35 Billion Tonnes

The long arc of annual global CO₂ emissions (1750–2024)



Reference visual: figure_3.jpg | Source: Our World in Data (Global Carbon Budget 2025)

Regional Share Shift, 1900 → 2024

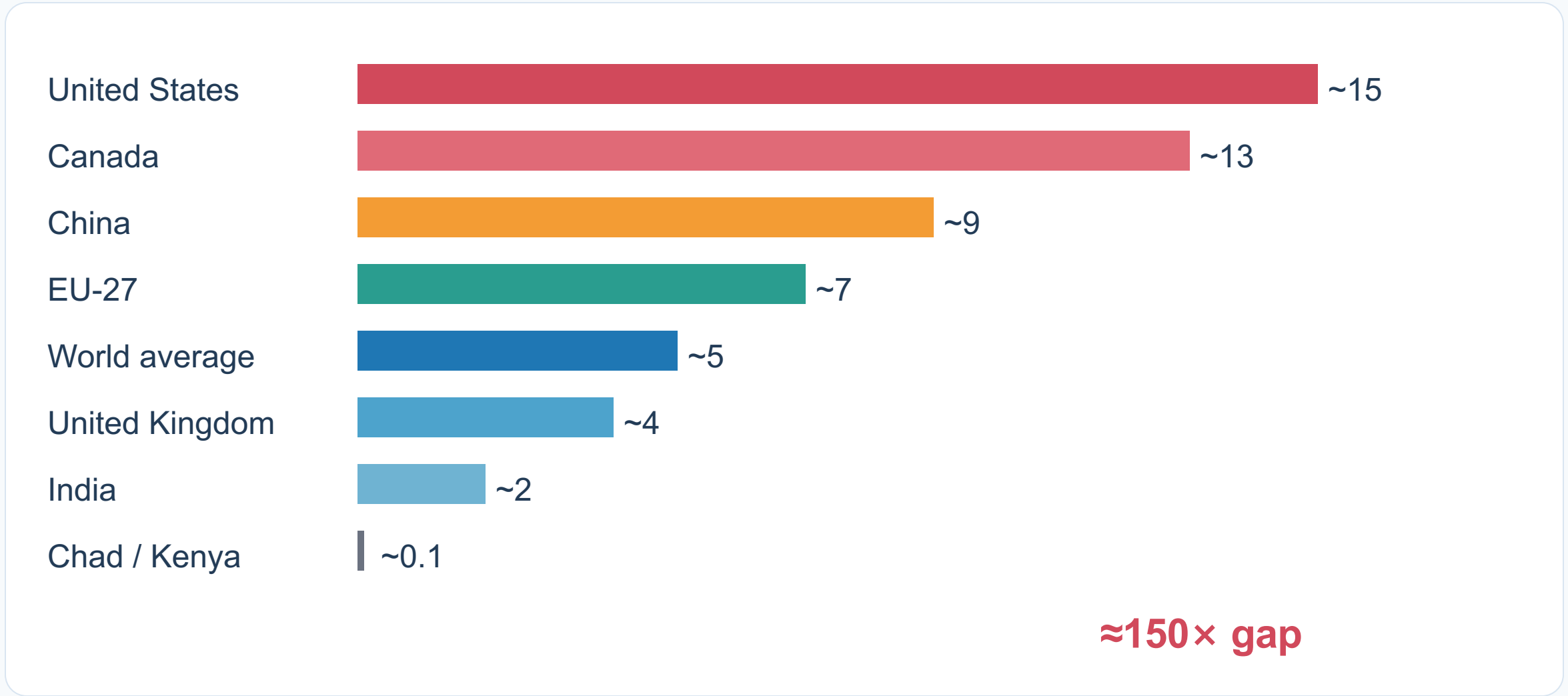
From Western dominance to Asian centrality



Source: source_brief regional shares; visual type requested: stacked comparison

Per Capita Emissions Reveal a 150× Inequality Gap

Tonnes CO₂ per person (selected economies)



2024 Annual Change in Emissions

Declines in US/Europe; growth across many emerging regions

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data

Table

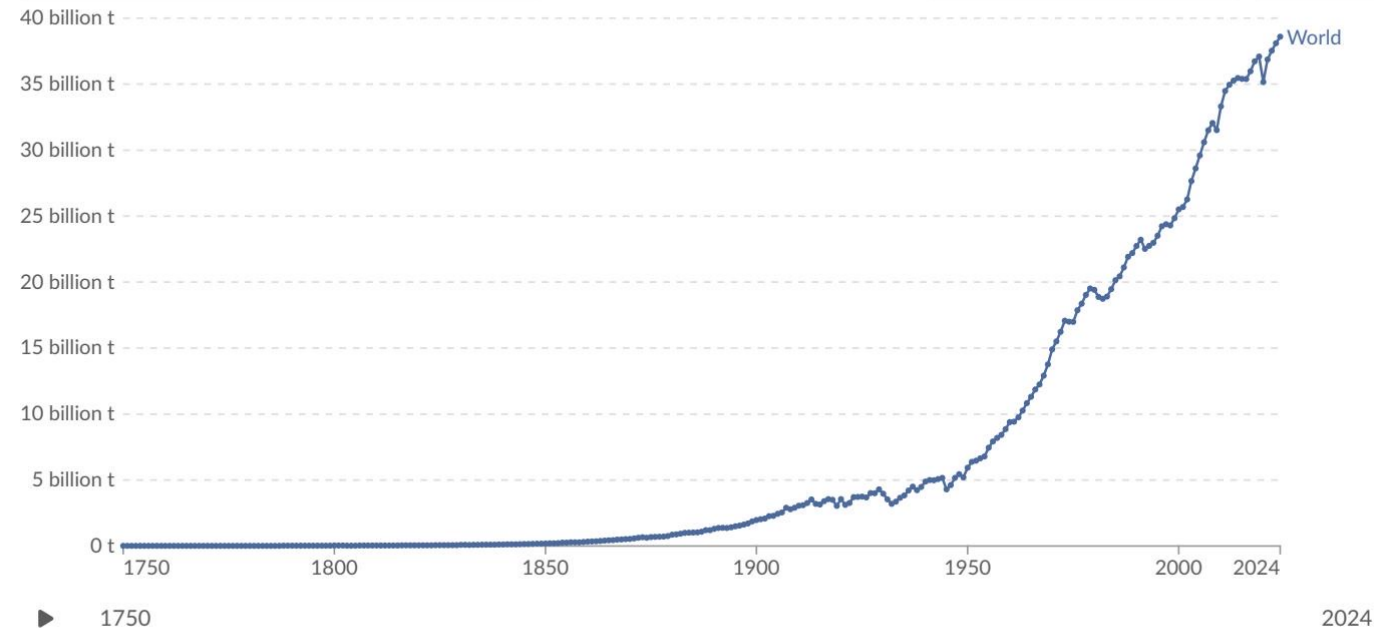
Map

Line

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Edit countries and regions

Settings



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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Snapshot Signals

- US and much of Europe: –1% to –5%
- Russia, parts of SE Asia, several African countries: +2% to +10%+
- Some South American countries showed >10% increases

Net message: global emissions still rise despite declines in mature economies.

Why Per Capita Emissions Differ: Energy Mix Matters

Policy and power-system structure can outweigh income similarities

Core mechanism

- Higher nuclear/renewables share generally reduces power-sector CO₂ intensity.
- Fossil-heavy systems tend to produce higher per-capita emissions.
- Similar prosperity can yield different emissions outcomes.

Implication: technology + policy choices matter.

Comparison Table (Illustrative from brief)

Country pair	Energy mix profile	Per-capita CO ₂
France / UK	Higher nuclear & renewables	~4 t/person
Germany / Netherlands	Higher fossil-fuel share	Higher than ~4

Note: source brief provides qualitative comparison and explicit ~4 t/person level for France/UK.

Key Takeaways

Emissions are still rising, but responsibility is shifting

- 1) **Global total has not yet peaked**: annual CO₂ remains above 35 Bt.
- 2) **China is the largest emitter**: >25% of global total (~12.5 Bt).
- 3) **Per capita inequality remains extreme**: about 150× gap.
- 4) **US and Europe are declining annually**,
but historical cumulative responsibility remains large.
- 5) **Energy mix and policy choices are decisive**: prosperity alone.

Policy focus: accelerate low-carbon energy while addressing equity and development needs.